

Active Contour Segmentation for Boundary Detection

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1 Context

The objective of this project is to detect the boundary of objects in grayscale images using active contours, also known as snakes.

A first natural idea would be to use the image gradient. Indeed, object boundaries usually correspond to strong intensity variations, and the gradient highlights these high-frequency regions. However, a gradient image alone is not always enough. It gives local edge pixels, but it does not directly give a clean, closed and regular contour around the object.

This is why we use a snake model. The idea is to initialize a rough contour around the object and then let it evolve iteratively under two kinds of forces:

- internal forces, which regularize the shape of the contour;
- external forces, which attract the contour toward the image boundaries.

In this project, I focus not only on obtaining good-looking contours, but also on understanding why the method works on some shapes and fails on others. In particular, I compare a mostly convex shape, such as a water droplet, with less convex or more difficult shapes, such as an hourglass, a heart or a puzzle piece.

The main goal is to study:

- the mathematical mechanism of active contours;
- the attraction of the snake toward image edges;
- the influence of the parameters;
- the robustness to small perturbations;
- and the limitations of the method on complex shapes.

2 Principle of Active Contours

A snake is a closed parametric curve:

$$\mathbf{c}(s) = \begin{bmatrix} x(s) \\ y(s) \end{bmatrix}, \quad s \in [0, 1].$$

Since the contour is closed:

$$\mathbf{c}(0) = \mathbf{c}(1).$$

In the implementation, the contour is discretized into K points:

$$\mathbf{c}_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix}, \quad i = 1, \dots, K.$$

The number of points K is important. If the number of points is too small, the snake cannot follow local details. If it is too large, the contour can become more sensitive to small variations and artifacts.

The snake evolves by minimizing an energy. The intuition is simple: the contour must be attracted by the image boundaries, but it must also remain geometrically coherent.

In practice, the first step is therefore to define an initial contour. For simple objects such as the water droplet, this initial contour can be a circle surrounding the object. This rough contour is not expected to be precise at the beginning. It only gives the snake a starting point from which it can progressively move toward the real boundary.

3 Mathematical Formulation

3.1 Snake Energy

The total snake energy can be written as:

$$E(\mathbf{c}) = E_{\text{internal}}(\mathbf{c}) + E_{\text{external}}(\mathbf{c}).$$

A classical formulation is:

$$E(\mathbf{c}) = \int_0^1 \left[\frac{\alpha}{2} \|\mathbf{c}'(s)\|^2 + \frac{\beta}{2} \|\mathbf{c}''(s)\|^2 + P(\mathbf{c}(s)) \right] ds.$$

Here:

- α controls the elasticity of the contour;
- β controls the curvature regularization;
- P is an image potential that attracts the snake toward object boundaries.

The first derivative term prevents the contour from stretching too freely. The second derivative term penalizes curvature and makes the contour smoother.

Therefore:

- high α makes the snake more tense;
- high β makes the snake smoother and less able to bend;
- low α and low β make the snake more flexible.

3.2 Classical Edge Potential

In the classical gradient-based snake, the image potential is often defined as:

$$P(x, y) = -\|\nabla(G_\sigma * I)(x, y)\|^2,$$

where:

- I is the input image;
- $G_\sigma * I$ is the image smoothed by a Gaussian filter;
- $\nabla(G_\sigma * I)$ is the gradient of the smoothed image.

The minus sign is important. At object boundaries, the gradient magnitude is high, so P becomes low. Since the snake minimizes the total energy, the contour is attracted toward high-gradient regions.

In the code, the strength of this attraction is not included inside P directly. It is controlled separately by the parameter γ .

3.3 Euler-Lagrange Equation

Let:

$$F(\mathbf{c}, \mathbf{c}', \mathbf{c}'') = \frac{\alpha}{2} \|\mathbf{c}'(s)\|^2 + \frac{\beta}{2} \|\mathbf{c}''(s)\|^2 + P(\mathbf{c}(s)).$$

The Euler-Lagrange equation associated with this second-order functional is:

$$\nabla P(\mathbf{c}(s)) - \alpha \mathbf{c}''(s) + \beta \mathbf{c}^{(4)}(s) = 0.$$

Equivalently:

$$\alpha \mathbf{c}''(s) - \beta \mathbf{c}^{(4)}(s) - \nabla P(\mathbf{c}(s)) = 0.$$

This equation describes a stationary contour. In practice, we do not solve it directly. Instead, we introduce an artificial evolution time τ and make the contour move progressively.

3.4 Evolution Equation

The gradient descent evolution can be written as:

$$\frac{\partial \mathbf{c}}{\partial \tau} = \alpha \mathbf{c}''(s) - \beta \mathbf{c}^{(4)}(s) + \gamma F_{\text{ext}}(\mathbf{c}(s)).$$

In this expression:

- $\alpha \mathbf{c}''(s)$ and $-\beta \mathbf{c}^{(4)}(s)$ are internal regularization terms;
- F_{ext} is the external force field;
- γ controls the strength of the external attraction.

For the classical edge-based snake:

$$F_{\text{ext}}(x, y) = -\nabla P(x, y) = \nabla \left(\|\nabla (G_\sigma * I)(x, y)\|^2 \right).$$

So the snake is pushed toward locations where the edge strength increases.

4 Discrete Scheme Used in the Code

In the code, the contour coordinates are represented by two vectors:

$$x = (x_1, \dots, x_K), \quad y = (y_1, \dots, y_K).$$

Since the contour is closed, the derivative matrices use cyclic boundary conditions. We define:

$$D = \alpha D_2 - \beta D_4,$$

where:

- D_2 is the discrete second derivative matrix;
- D_4 is the discrete fourth derivative matrix.

The spatial discretization step along the contour is not written explicitly here. In practice, its effect is absorbed into the empirical parameters α and β .

The update used in the code is semi-implicit:

$$\begin{cases} x^{n+1} = A(x^n + \Delta t \gamma F_x(x^n, y^n)), \\ y^{n+1} = A(y^n + \Delta t \gamma F_y(x^n, y^n)), \end{cases}$$

with:

$$A = (I - \Delta t D)^{-1}.$$

This is not a pure explicit Euler scheme. The internal regularization part is handled implicitly through the matrix A , while the external force is evaluated explicitly at the current contour position.

This is useful because the internal regularization can be stiff, especially when β is large.

5 External Forces in the Current Project

The current code supports several external force modes:

- `edge`;
- `distance`;
- `mixed`;
- `mask_distance`.

5.1 Edge Mode

The **edge** mode is the closest to the classical snake. The image is smoothed, an edge map is computed, and the snake is attracted by the gradient of this edge map.

This mode is useful to understand the basic behaviour of snakes, but it has a limited capture range. If the initial contour is too far from the object boundary, the attraction can be weak.

5.2 Distance Mode

The **distance** mode first extracts strong edges from the edge map and then computes a distance transform to these edges.

If $d_{\partial\Omega}(x, y)$ denotes the distance from point (x, y) to the nearest detected boundary, the external force is defined in the direction of decreasing distance:

$$F_{\text{ext}}(x, y) = -\frac{\nabla d_{\partial\Omega}(x, y)}{\|\nabla d_{\partial\Omega}(x, y)\| + \varepsilon}.$$

In the implementation, this normalization makes the force mainly directional. The strength of the attraction is then controlled separately by the parameter γ .

This force points toward the closest detected boundary. It gives a larger capture range than the raw gradient.

5.3 Mask Distance Mode

The **mask_distance** mode is the main mode used in the experiments. It is useful when the object has a clear contrast with the background.

The pipeline is:

1. threshold the image using the selected polarity;
2. keep the largest connected foreground component;
3. fill holes inside the object;
4. extract the external boundary of this foreground mask;
5. compute the distance transform to this boundary;
6. use the normalized negative distance gradient as external force.

The force is:

$$F_{\text{ext}}(x, y) = -\frac{\nabla d_{\partial\Omega}(x, y)}{\|\nabla d_{\partial\Omega}(x, y)\| + \varepsilon},$$

where $\partial\Omega$ is the external boundary extracted from the estimated object mask.

Therefore, the experiments in this report should be understood as an active contour pipeline using a mask-based distance force, rather than a purely gradient-based classical snake.

This mode is especially useful for images where the object is dark on a bright background or bright on a dark background.

5.4 Mixed Mode

The **mixed** mode combines edge attraction and distance attraction. It can be useful when we want both a global attraction toward the boundary and local edge information.

However, if the image has many competing edges, it can become less stable.

6 Influence of Parameters

The behaviour of the snake depends strongly on the parameters.

6.1 α : Elasticity

The parameter α controls the tension of the contour.

If α is high, the contour tends to shrink and stay regular. This is useful for smooth convex shapes, but it can prevent the snake from following local details.

If α is low, the snake becomes more flexible.

6.2 β : Curvature

The parameter β controls the curvature penalty.

A high β produces a smooth contour. This is good for a water droplet, but it can be problematic for shapes with concavities.

If β is too high, the snake may cut concave regions because it prefers a shorter and smoother contour.

6.3 γ : External Attraction

The parameter γ controls the strength of the attraction toward the external force field.

If γ is too low, the contour may not reach the object boundary. If γ is too high, the snake may react too strongly to small artifacts.

6.4 Δt : Time Step

The time step Δt controls how much the contour moves at each iteration.

A large Δt can make the snake unstable or oscillating. A smaller value gives a more stable evolution but may require more iterations.

6.5 Number of Points

The parameter `n_points` controls the discretization of the contour.

Simple smooth shapes can be represented with fewer points. More complex shapes require more points to capture local variations.

6.6 Contour Smoothing

The code contains a parameter called `contour_smoothing_sigma`. In the experiments presented in this report, this value is set to zero.

Therefore, no additional Gaussian smoothing is applied to the contour during the evolution. The geometric regularization comes from the snake internal energy controlled by α and β .

7 Project Pipeline

The implemented pipeline is:

1. load the input image;
2. convert it to grayscale;
3. normalize intensities between 0 and 1;

4. select the object polarity: **dark**, **bright** or **auto**;
5. initialize a contour around the object;
6. build the external force field;
7. evolve the contour with the semi-implicit update;
8. save the final contour and visual outputs.

For each experiment, the generated output includes:

- the initial contour;
- the final contour;
- the snake evolution;
- the edge or boundary attraction map;
- the final contour coordinates;
- the configuration file.

8 Parameters Used in the Experiments

The following table summarizes the main parameter values used in the experiments.

Shape	Force mode	α	β	γ	Δt	Points	Iterations
Droplet	mask_distance	0.20	2.00	1.00	1.00	100	1200
Hourglass	mask_distance	0.08	0.45	1.60	0.80	160	1200
Hourglass + filaments	mask_distance	0.08	0.45	1.60	0.80	160	1200
Heart	mask_distance	0.06	0.30	1.80	0.70	180	1600
Puzzle piece	mask_distance	0.05	0.25	1.80	0.60	200	500

Table 1: Main parameters used for the active contour experiments. In all experiments, `contour_smoothing_sigma` is set to zero.

The droplet uses higher regularization because its boundary is smooth and mostly convex. The hourglass, heart and puzzle piece use lower values of α and β because they require more flexibility.

9 Experiments

9.1 Convex Case: Water Droplet

The first experiment is performed on a water droplet. This is a favourable case because the object is mostly convex and its boundary is smooth.

For this image, the initial snake is created as a rough contour around the object. In the simplest interpretation, it can be seen as a circular initialization enclosing the droplet. Then, during the iterations, the contour progressively moves toward the real boundary.

This figure is useful because it shows the actual evolution of the snake from the initial contour to the final contour. We can see that the contour starts as a coarse approximation and is progressively attracted by the external boundary of the object.

Snake evolution



Figure 1: Evolution of the snake on the water droplet. The initial contour is a rough contour around the object, and the snake progressively converges toward the external boundary.

This result illustrates a case where active contours are well adapted. The internal regularization helps produce a clean contour without preventing the snake from reaching the object boundary.

9.2 Less Convex Case: Hourglass

The hourglass is more difficult than the droplet. It is less convex and contains a narrow central region. Therefore, the contour must deform more locally.

For this reason, α and β are reduced compared to the droplet. The number of points is also increased.

Final contour



Figure 2: Final contour on the hourglass. The object is less convex, so the snake requires lower regularization.

This experiment shows that the same method can work on less convex shapes, but only if the parameters are adapted.

10 Robustness to Small Black Filaments

To test robustness, small black filaments were added around the hourglass. The goal was to see whether the snake would follow these small external perturbations or still detect the main object boundary.

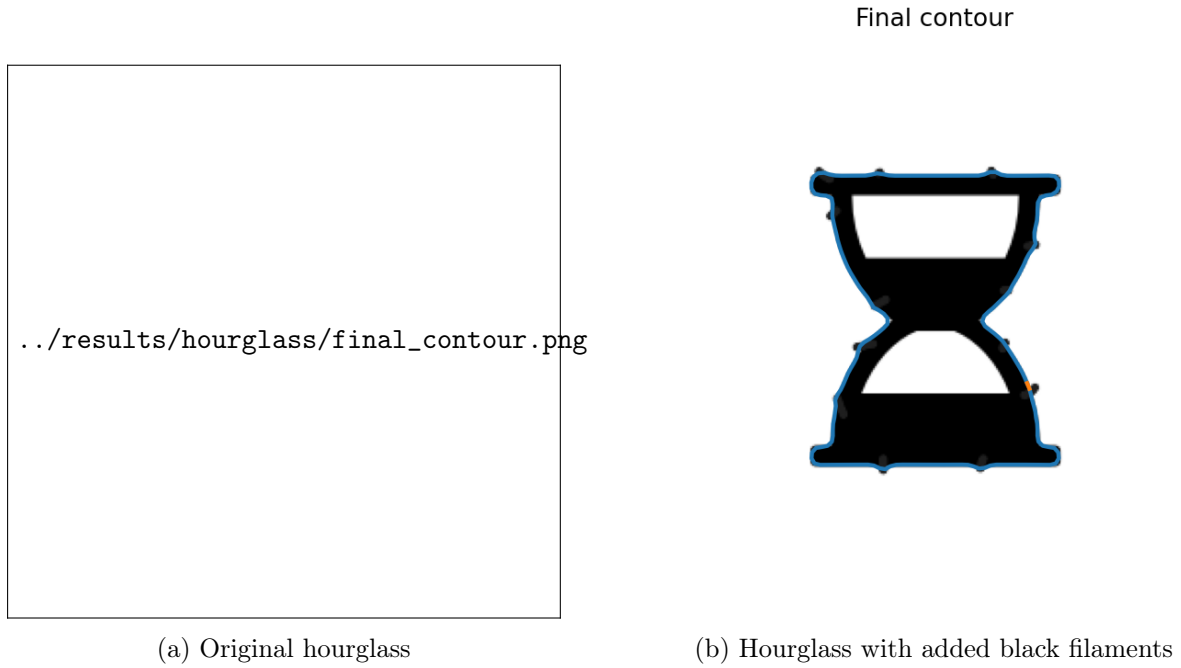


Figure 3: Comparison between the original hourglass and the modified hourglass with black filaments.

In this experiment, the final contour remains close to the original prediction. This suggests that the dominant object boundary has more influence than the small added perturbations, at least for this configuration.

This behaviour is due to the balance between the external attraction and the internal regularization. The snake does not only follow isolated pixels; it also tries to keep a coherent closed shape.

11 Other Shapes

11.1 Heart

The heart is an intermediate case. It has a clear global boundary, but also a concavity at the top.

With lower values of α and β , the snake becomes more flexible and can better adapt to the shape. However, very deep concavities can still be difficult.

Final contour



Figure 4: Final contour on the heart image.

11.2 Puzzle Piece

The puzzle piece is one of the most difficult cases. It contains external bumps and concave regions.

The snake can follow the global contour and some external bumps, but it may cut some concave parts. This is not only a tuning problem. It comes from the structure of the snake energy itself.

Final contour

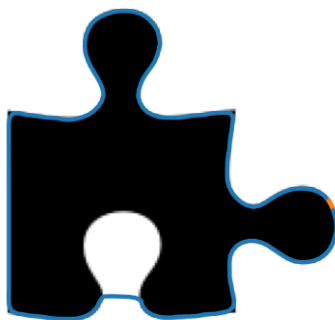


Figure 5: Final contour on the puzzle piece. The snake detects the global shape but may cut some concave regions.

Entering a deep concavity increases both the contour length and the curvature. Since the snake penalizes these quantities, it can prefer a smoother shortcut.

11.3 Additional Qualitative Cases

Other images such as the hand, the leaf and the champagne glass are useful to understand the limits of the method.

The hand is difficult because of the concavities between fingers. The champagne glass is difficult because it is thin and elongated. The leaf can be easier if the external boundary is strong, but internal texture may create competing edges.

These cases show that active contours are very dependent on object geometry, contrast and initialization.

12 Discussion

The experiments show that active contours are useful, but not universal.

They work well when:

- the object is well separated from the background;
- the boundary is visible;
- the shape is smooth or mostly convex;
- the initialization surrounds the object correctly;
- the parameters are adapted to the geometry.

They become more difficult when:

- the object has strong concavities;
- the background contains many edges;
- the object boundary is weak;
- the object is thin or fragmented;
- the initialization is too far from the real boundary.

A simple gradient can reveal edge pixels, but it does not directly provide a clean closed contour. The snake adds geometric regularization, which is useful for smooth objects. But this same regularization can become a limitation for highly concave objects.

This explains the difference between the droplet and the puzzle piece. The droplet matches the natural bias of the snake. The puzzle piece does not, because some parts require the contour to bend strongly.

13 Possible Improvements

13.1 Gradient Vector Flow

A strong improvement would be to use Gradient Vector Flow. GVF diffuses the gradient field into homogeneous regions and improves the capture range of the snake.

This could help the contour enter concave regions, especially for shapes like the puzzle piece or the hand.

13.2 Balloon Force

Another improvement would be to add a balloon force. A balloon force can push the contour outward or inward, which can help when the snake is initialized inside or outside the object.

13.3 Comparison with Canny

It would also be interesting to compare the snake with Canny edge detection. Canny provides an edge map, while the snake provides a closed and regular contour.

13.4 Quantitative Evaluation

If manually annotated masks are available, the results could be evaluated using:

- Intersection over Union;
- Dice score;
- Hausdorff distance;
- mean contour distance.

14 Conclusion

In this project, I implemented and studied active contour segmentation for boundary detection.

The method is based on a balance between internal regularization and external attraction toward image boundaries. The mathematical formulation explains why the snake naturally prefers smooth contours and why it is attracted toward object edges.

The experiments show that the method works well on smooth and mostly convex objects such as the water droplet. It also works on less convex objects such as the hourglass, but the parameter tuning becomes more important.

The robustness test with black filaments around the hourglass shows that small perturbations do not necessarily change the final contour when the main object boundary remains dominant.

However, the method has clear limitations. For highly concave objects such as a puzzle piece or a hand, the snake may cut concave regions because it prefers shorter and smoother contours.

Overall, this project shows both the strengths and the limits of active contour segmentation. It is a useful image processing method for understanding the interaction between edge attraction and geometric regularization.